

EE 210L: Digital Circuits and Logic Design Lab

Lab 7: Shift Registers and Counters

Name: _____ Date: _____

Partner(s): _____ Section: _____

The 74LS73 uses V_{CC} on pin 4 / GND on pin 11, and the 74LS90 uses V_{CC} on pin 5 / GND on pin 10. Confirm both before applying power.

Part 5.1: Ripple Counter

Signal	Measured frequency	Ratio to clock
Clock		1
1Q (stage 1)		
2Q (stage 2)		

Paste the capture showing that 2Q changes after 1Q rather than with it.



Figure 1: Ripple delay between counter stages.

Measured delay from the 1Q edge to the 2Q edge: _____ ns

Estimated worst-case delay for an 8-bit ripple counter built the same way: _____ ns

Part 5.2: Synchronous Counter (74LS163)

Count	QD	QC	QB	QA	Decimal	RCO
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						
13						
14						
15						
16						

RCO asserts at state _____. Output frequencies at 10 kHz clock: QA = _____, QD = _____

Paste the capture showing QA and QD relative to the clock at 10 kHz.

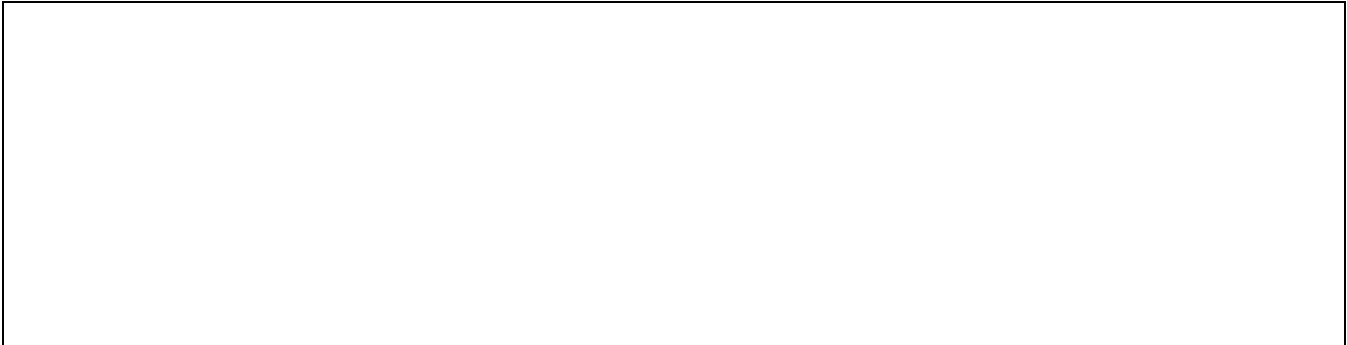


Figure 2: Synchronous counter output frequencies (QA and QD) versus the clock.

Part 5.3: Mod-10 Counter

Outputs NAnDED to detect state 9: _____ and _____

Count	QD	QC	QB	QA	Decimal
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					

Why does state 10 never appear, even briefly? What would be different on a counter with an asynchronous clear?

Part 5.4: Decade Counter (74LS90)

Count	QD	QC	QB	QA	Decimal
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					

Effect of taking R0(1) and R0(2) HIGH together:

Part 5.5: Universal Shift Register (74LS194)

5.5.1 Parallel Load and Hold

Mode (S1 S0)	After clock edge	QA	QB	QC	QD
1 1 (load 1011)	1st				
0 0 (hold)	after several				

5.5.2 Shift Right (S1 S0 = 01, SR SER = 0)

Clock edge	QA	QB	QC	QD
start				
1				
2				
3				
4				

5.5.3 Shift Left (S1 S0 = 10, SL SER = 0)

Clock edge	QA	QB	QC	QD
start				
1				
2				
3				
4				

Part 5.6: Ring Counter

Clock edge	QA	QB	QC	QD
start (loaded 1000)				
1				
2				
3				
4				
5				
6				
7				
8				

Part 5.7: Johnson Counter

Clock edge	QA	QB	QC	QD
start (cleared)				
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

5.7.1 States per Flip-Flop

Counter type	States from 4 flip-flops	Decoding logic needed per state
Ring		
Johnson		
Binary		

Discussion and Conclusions

Address at least: what decoding glitches are and why a ripple counter produces them, why a synchronous clear matters for the mod-10 design, and the trade-off between ring, Johnson, and binary counters.
